

PROJECT — COURIER DELIVERY MANAGEMENT SYSTEM

1. Context

Một công ty giao hàng quy mô nhỏ đến trung bình cần một hệ thống máy tính đơn giản để quản lý thông tin khách hàng, bưu kiện, đơn giao hàng, nhân viên giao nhận và thanh toán phí vận chuyển. Hệ thống phải lưu trữ dữ liệu chi tiết về khách hàng như mã khách hàng, họ tên, số điện thoại, địa chỉ; đồng thời quản lý thông tin bưu kiện như mã kiện hàng, loại hàng, trọng lượng, địa chỉ gửi, địa chỉ nhận, phí vận chuyển và trạng thái giao hàng.

Nhân viên điều phối cần có thể tạo đơn giao hàng, cập nhật thông tin khách hàng, phân công nhân viên giao nhận, theo dõi trạng thái đơn hàng và tính phí vận chuyển. Hệ thống cũng phải hỗ trợ xem lịch sử giao hàng, danh sách đơn đang xử lý và báo cáo doanh thu hoặc các nhân viên giao hàng có số đơn hoàn thành cao nhất.

Mục tiêu của hệ thống là giúp công ty giao hàng quản lý đơn vận chuyển chính xác hơn, giảm sai sót trong quá trình xử lý đơn và hỗ trợ vận hành hàng ngày hiệu quả hơn.

2. Users

- **Reception Staff** – tiếp nhận khách hàng, tạo đơn giao hàng và cập nhật thông tin bưu kiện.
 - **Dispatcher** – phân công shipper và theo dõi trạng thái giao hàng.
 - **Delivery Staff** – cập nhật quá trình giao nhận.
 - **Manager** – xem báo cáo doanh thu và hiệu suất giao hàng.
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3. Functional Requirements

Customer Management

1. Add new customer with details (ID, name, phone number, address, ...).
2. Update customer information.
3. Remove a customer.
4. View all customers.
5. Search customers by name or phone number.

Parcel Management

1. Add new parcel with details (ID, sender ID, receiver name, receiver phone, pickup address, delivery address, weight, type, ...).
2. Update parcel information.
3. Remove a parcel.
4. View all parcels.
5. Search parcels by ID, receiver name, or status.

Delivery Order Management

1. Create a delivery order for a parcel.
2. Update delivery order information.
3. Cancel a delivery order.
4. View delivery order details.
5. View delivery history of a customer.

Staff Management

1. Add delivery staff information.
2. Update delivery staff information.
3. Assign delivery staff to an order.
4. View assigned orders of a delivery staff.
5. Search delivery staff by name or phone number.

Delivery Tracking Management

1. Mark order status as Pending, Picked Up, In Transit, Delivered, Failed, or Cancelled.
2. Update delivery note.
3. Record pickup date and delivery date.
4. View orders currently in transit.
5. List failed delivery orders.

Billing Management

1. Calculate delivery fee.
2. Add extra fee for urgent delivery if applicable.
3. View invoice details.
4. Record payment.
5. Generate final invoice.

Reporting

1. Generate daily revenue report.
2. Generate monthly revenue report.
3. List most active delivery staff.
4. List successfully delivered orders.
5. List failed delivery orders.

4. Business Rules

BR1: Each Customer ID must be unique in the system and cannot be modified after creation.

BR2: Each Parcel ID must be unique in the system and cannot be modified after creation.

BR3: Each Delivery Order ID must be unique in the system and cannot be modified after creation.

BR4: Customer name and phone number must not be empty.

BR5: Receiver name, receiver phone, pickup address, and delivery address must not be empty.

BR6: Parcel weight must be greater than 0.

BR7: A customer must exist before a parcel can be created.

BR8: A parcel must exist before a delivery order can be created.

BR9: A delivery staff member must exist and be active before being assigned to an order.

BR10: An order can only be assigned to one delivery staff member at a time.

BR11: Order status must be one of: Pending, Picked Up, In Transit, Delivered, Failed, or Cancelled.

BR12: A cancelled order cannot be delivered or billed.

BR13: Only delivered orders can be finalized for payment.

BR14: Delivery fee is calculated based on parcel weight and delivery type.

BR15: An urgent delivery may have an additional predefined fee.

BR16: Pickup date must be earlier than or equal to delivery date for completed orders.

BR17: Payment status must be one of: Unpaid, Partially Paid, or Paid.

BR18: All user input must be validated before processing.

BR19: Most active delivery staff are determined based on the number of delivered orders in a selected period.

BR20: Revenue reports are calculated from paid invoices only.

5. OOP Requirements

- **Encapsulation**

Ví dụ: `Customer`, `Parcel`, `DeliveryOrder`, `DeliveryStaff`, `Invoice` classes.

- **Inheritance**

Ví dụ: `Parcel` → `DocumentParcel`, `GoodsParcel`.

- **Polymorphism**

Ví dụ: override method `calculateFee()` cho từng loại kiện hàng.

- **Collections**

Sử dụng `List`, `Map` để lưu khách hàng, kiện hàng, đơn giao, nhân viên và hóa đơn.

- **Exception handling**

Xử lý lỗi nhập sai dữ liệu, mã không tồn tại, trạng thái không hợp lệ, phân công trùng, trọng lượng không hợp lệ,...

- **Class diagram before coding**

6. Console UI Requirements

Menu-based interface:

- Manage Customers
- Manage Parcels
- Delivery Order Management
- Manage Delivery Staff
- Delivery Tracking Management
- Billing Management
- Reports
- Exit

Yêu cầu:

- Validate user input và validate business rules.
- Use console I/O and File I/O.
- Display messages (Success / Fail) for each task.

PROJECT — COURIER DELIVERY MANAGEMENT SYSTEM

1. MAIN MENU

```
=====
      COURIER DELIVERY MANAGEMENT
=====
1. Manage Customers
2. Manage Parcels
3. Delivery Order Management
4. Manage Delivery Staff
5. Delivery Tracking Management
6. Billing Management
7. Reports
8. Exit
-----
Choose an option: _
```

CUSTOMER MANAGEMENT

Task B1 — Add Customer

GUI

```
----- ADD CUSTOMER -----  
Customer ID: C01  
Full Name: Nguyen Van Phuc  
Phone Number: 0988123456  
Address: Da Nang  
[1] Save [2] Cancel
```

Output

Customer added successfully.

Task B2 — Update Customer

GUI

```
----- UPDATE CUSTOMER -----  
Enter Customer ID to update: C01  
Current Information:  
Name: Nguyen Van Phuc  
Phone Number: 0988123456  
Address: Da Nang  
  
Enter new Phone Number (leave blank to skip): 0911222333  
Enter new Address (leave blank to skip): Hoi An  
[1] Update [2] Cancel
```

Output

Customer updated successfully.

Task B3 — View All Customers

```
----- CUSTOMER LIST -----  
ID      Name           Phone Number   Address  
-----  
C01     Nguyen Van Phuc    0911222333    Hoi An  
C02     Tran Thi Mai       0977666555    Da Nang  
-----  
Press ENTER to return...
```

PARCEL MANAGEMENT

Task B4 — Add Parcel

GUI

```
----- ADD PARCEL -----  
Parcel ID: P01  
Sender ID: C01  
Receiver Name: Le Minh Anh  
Receiver Phone: 0909888777  
Pickup Address: Da Nang  
Delivery Address: Hue  
Weight: 2.5  
Type: Goods  
[1] Save [2] Cancel
```

Output

Parcel added successfully.

Task B5 — View All Parcels

```
----- PARCEL LIST -----  
ID      Sender ID  Receiver Name  Weight  Type      Status  
-----  
P01     C01           Le Minh Anh   2.5 kg  Goods     Pending  
P02     C02           Pham Thu Ha   0.5 kg  Document  Delivered  
-----  
Press ENTER to return...
```

DELIVERY ORDER MANAGEMENT

Task B6 — Create Delivery Order

GUI

```
----- CREATE DELIVERY ORDER -----  
Order ID: O01  
Parcel ID: P01  
Order Date: 25/04/2026  
Delivery Type: Standard  
Expected Delivery Date: 27/04/2026  
Status: Pending
```

Output

Delivery order created successfully.

Task B7 — Update Delivery Order

GUI

```
----- UPDATE DELIVERY ORDER -----  
Enter Order ID to update: 001  
Current Information:  
Parcel ID: P01  
Delivery Type: Standard  
Expected Delivery Date: 27/04/2026  
  
Enter new Delivery Type (leave blank to skip): Urgent  
Enter new Expected Delivery Date (leave blank to skip): 26/04/2026  
[1] Update [2] Cancel
```

Output

Delivery order updated successfully.

Task B8 — View Delivery Order Details

```
----- ORDER DETAILS -----  
Order ID: 001  
Parcel ID: P01  
Sender: Nguyen Van Phuc  
Receiver: Le Minh Anh  
Pickup Address: Da Nang  
Delivery Address: Hue  
Delivery Type: Urgent  
Expected Delivery Date: 26/04/2026  
Status: Pending  
-----  
Press ENTER to return...
```

DELIVERY STAFF MANAGEMENT

Task B9 — Add Delivery Staff

GUI

```
----- ADD DELIVERY STAFF -----  
Staff ID: S01  
Full Name: Tran Minh Duc  
Phone Number: 0933444555  
Vehicle Type: Motorbike  
Status: Active  
[1] Save [2] Cancel
```

Output

Delivery staff added successfully.

Task B10 — Assign Delivery Staff

GUI

```
----- ASSIGN DELIVERY STAFF -----  
Order ID: O01  
Staff ID: S01  
Assignment Date: 25/04/2026
```

Output

Delivery staff assigned successfully.

Task B11 — View Assigned Orders of Staff

```
----- ASSIGNED ORDERS -----  
Staff ID: S01  
-----  
Order ID   Parcel ID   Receiver Name   Status  
-----  
O01        P01          Le Minh Anh     Pending  
O05        P07          Nguyen Bao An   In Transit  
-----  
Press ENTER to return...
```

DELIVERY TRACKING MANAGEMENT

Task B12 — Update Order Status

GUI

```
----- UPDATE ORDER STATUS -----  
Order ID: O01  
Current Status: Pending  
New Status: Picked Up
```

Output

Order status updated successfully.

Task B13 — Add Delivery Note

GUI

```
----- ADD DELIVERY NOTE -----  
Order ID: 001  
Note: Parcel picked up from sender at 09:15 AM.
```

Output

Delivery note added successfully.

Task B14 — View Orders in Transit

```
----- ORDERS IN TRANSIT -----  
Order ID   Receiver Name   Staff Name   Status  
-----  
001        Le Minh Anh     Tran Minh Duc   In Transit  
008        Pham Thu Ha     Le Van Nam     In Transit  
-----  
Press ENTER to return...
```

Task B15 — View Failed Orders

```
----- FAILED DELIVERY ORDERS -----  
012 - Receiver unavailable - 24/04/2026  
015 - Wrong delivery address - 25/04/2026  
-----  
Press ENTER to return...
```

BILLING MANAGEMENT

Task B16 — Calculate Invoice

GUI

```
----- CALCULATE INVOICE -----  
Order ID: 001  
Base Fee: 40,000  
Urgent Charge: 20,000
```

Output

Invoice calculated successfully.
Base Fee: 40,000 VND

Urgent Charge: 20,000 VND
Total Amount: 60,000 VND

Task B17 — View Invoice Details

GUI

```
----- INVOICE DETAILS -----  
Invoice ID: INV01  
Order ID: 001  
Sender: Nguyen Van Phuc  
Receiver: Le Minh Anh  
-----  
Base Fee:           40,000 VND  
Urgent Charge:      20,000 VND  
Total Amount:       60,000 VND  
Payment Status:     Unpaid  
-----  
Press ENTER to return...
```

Task B18 — Record Payment

GUI

```
----- RECORD PAYMENT -----  
Invoice ID: INV01  
Amount Paid: 60,000  
Payment Method: Cash  
Payment Date: 26/04/2026
```

Output

Payment recorded successfully.
Invoice status updated to Paid.

REPORTS

Task B19 — Daily Revenue Report

```
----- DAILY REVENUE REPORT -----  
Date: 26/04/2026  
Total Paid Invoices: 22  
Total Delivered Orders: 22  
Total Revenue: 1,320,000 VND  
-----  
Press ENTER to return...
```

Task B20 — Monthly Revenue Report

```
----- MONTHLY REVENUE REPORT -----  
Month/Year: 04/2026  
Total Paid Invoices: 465  
Total Revenue: 28,750,000 VND  
-----  
Press ENTER to return...
```

Task B21 — Most Active Delivery Staff

```
----- MOST ACTIVE DELIVERY STAFF -----  
S01 Tran Minh Duc   - 128 delivered orders  
S03 Le Van Nam      - 116 delivered orders  
S05 Pham Quoc Bao   - 104 delivered orders  
-----  
Press ENTER to return...
```

Task B22 — Successfully Delivered Orders

```
----- SUCCESSFULLY DELIVERED ORDERS -----  
O01 - Le Minh Anh - 26/04/2026  
O02 - Pham Thu Ha - 26/04/2026  
O03 - Nguyen Bao An - 26/04/2026  
-----  
Press ENTER to return...
```